Spatial Distribution of Bidirectional Charging Stations by Simulation and Evaluation of Traffic and Grid Data

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Abstract. This papers goal is the investigation on how bidirectional charging stations can be distributed across a dynamic area by combining traffic simulation with partially extracted grid data from official sources. We model charging demand as a spatiotemporal process and evaluate candidate locations with respect to accessibility, traffic flow, and waiting time. The proposed workflow integrates mobility traces, station utilization forecasts, and synthetic grid constraints to identify robust deployment strategies. Results indicate that jointly optimizing traffic and grid criteria can reduce peak grid load while improving charging availability in high-demand districts.

Keywords: Bidirectional charging, traffic simulation, SUMO, OpenStreetMap, spatio-temporal analysis, charging demand, V2G

Nomenclature

BEAM	Behavior, Energy, Autonomy, and Mobility
EV	Electric Vehicle
MATSim	Multi-Agent Transport Simulation
POI	Point of Interest
SoC	State of Charge
SUMO	Simulation of Urban Mobility
TraCI	Traffic Control Interface
V2G	Vehicle-to-Grid
OSM	OpenStreetMap

1. Introduction

The rapid growth of electric mobility is shifting charging-infrastructure planning from static placement rules toward data-driven decision support [1]. Bidirectional charging adds a further dimension: vehicles are no longer only energy consumers but can also feed power back to the grid, a concept known as Vehicle-to-Grid (V2G), turning charging points into distributed flexibility assets [2]. Deciding where to place such stations therefore requires understanding where and when charging demand actually concentrates.

Because electric vehicles still represent a small share of the fleet, there is little real-world data on where charging stations should be placed and how intensively they are used. At the same time, uncoordinated charging concentrates electrical load in space and time and can stress local distribution grids [3], which makes the spatial and temporal structure of demand a central planning question. Existing siting studies, however, often rely on data or tooling that is not openly reproducible [4]. We therefore ask: *how can candidate locations for charging stations be ranked from openly available data, based on their accessibility and on how charging demand is distributed over space and time?*

To address this, we build a simulation workflow on openly available data. Road-network and land-use information are taken from OpenStreetMap (OSM), and traffic is simulated microscopically in Simulation of Urban Mobility (SUMO), controlled through its Python interface Traffic Control Interface (TraCI). From land-use Point of Interests (POIs)—residential buildings, offices, and amenities—we derive travel demand, and we model charging as a *spatiotemporal* process, that is, demand that varies simultaneously by location and time of day. The model distinguishes public charging stations from private home wallboxes with vehicle-specific access control. A custom web interface, which is inspired by SUMO's web interface, lets users select an area of interest on a map and generate ready-to-run scenarios directly, lowering the barrier to reproducing and extending the analysis. Candidate sites are then assessed in terms of accessibility, station utilization, and the resulting local charging load.

The scope of this study is the planning and evaluation stage. We focus on the spatial and temporal distribution of charging demand and load under realistic mobility patterns; a detailed electrical model of the distribution grid (power flow, feeder and transformer constraints) as well as battery degradation and market or pricing mechanisms are beyond its scope and left to future work.

1.1 Motivation

Urban charging demand is highly uneven across time and location, which can cause queues at popular stations while other sites remain virtually unused. At the same time, cars that are parked for extended periods of time (in residential or commercial areas) can remain plugged in, creating opportunities for bidirectional energy exchange. This means that the spatial distribution of charging stations can have a significant impact on both user experience and grid load. For example, placing stations in high-traffic areas may improve accessibility but could also lead to congestion and increased waiting times. Conversely, locating stations in less busy areas might reduce queues but could limit their utilization and the potential for vehicle-to-grid (V2G) services. Understanding the spatial and temporal patterns of charging demand is therefore crucial for effective infrastructure planning. However, there is a lack of open, reproducible methods for assessing candidate locations based on real-world mobility patterns and grid constraints.

1.2 Problem Statement

Given the afforementioned challenges, the problem statement of this paper can be formulated as follows: How can we determine the optimal spatial distribution of bidirectional charging stations, while regarding existing charging stations, in an urban area by integrating traffic simulation data with partially synthetic grid constraints, in order to maximize accessibility and minimize waiting times for electric vehicle users?

1.3 Contributions

This paper contributes a practical and reproducible workflow for planning bidirectional charging infrastructure with coupled traffic and grid evaluation. This work contains a user interface for deterministic scenario-based traffic generation along with a thorough analysis of candidate station portfolios under varying demand and grid conditions. By integrating synthetic grid constraints with mobility-based accessibility metrics, we provide a holistic framework for siting decisions that can be adapted to different urban contexts and policy objectives.

The main contributions of this work are as follows:

- A simulation-based siting pipeline that jointly evaluates mobility accessibility and distribution-grid impact.

- A multi-criteria scoring framework for comparing candidate station portfolios under proto-realistic demand variability.
- A reference implementation in Python to support reproducible experiments and scenario-based sensitivity studies.
- A reality-based implementation of synthetic wallbox profiles, for which no real-world data ist currently available.
- A fully working pipeline for independent scenario generation, traffic simulation, and evaluation of candidate station portfolios, which can be adapted to different urban contexts.
- A SUMO-inspired user interface for scenario-based traffic generation and evaluation of candidate station portfolios, enabling stakeholders to adjust parameters and visualize the impact of different siting strategies.

2. Related Work

Planning charging infrastructure has been studied extensively, and recent reviews organize the field along the dimensions of station siting, sizing, grid reinforcement, trip-considerations and the treatment of uncertainty [5, 13]. Most contributions optimize either where demand is best served or how the grid copes with the additional load; bidirectional charging, in which vehicles also feed power back, is only beginning to be integrated into such planning frameworks [9].

2.1 Traffic-Aware Siting

Traffic-aware siting methods estimate charging demand from origin-destination flows, activity patterns, and route choices, often placing stations so as to maximize coverage of traffic along routes. Data-driven analyses suggest that charging demand is largely shaped by proximity to amenities, Electric Vehicle (EV) density, and adjacency to high-traffic corridors, which motivates deriving demand from land-use data as we do [6]. Microscopic traffic simulators are increasingly used for this purpose: SUMO in particular provides built-in energy and charging-infrastructure models and has been used to assign electric vehicles to charging stations from simulated traffic data [7, 8].

A different approach than SUMO, is a mesoscopic traffic simulation model that uses a network of nodes and links to represent the road network, and simulates traffic flow based on traffic demand and road capacity. This approach can be used to estimate charging demand at a more aggregated level, and can be useful for planning charging infrastructure in areas with limited data availability or where detailed traffic simulation is not feasible. Multi-Agent Transport Simulation (MATSim) is an example of a mesoscopic traffic simulation model that has been used for this purpose, and paired with Behavior, Energy, Autonomy, and Mobility (BEAM) it is possible to simulate the impact of charging infrastructure on a dynamic traffic network [11, 12].

2.2 Grid-Aware Siting

Grid-aware planning approaches prioritize feeder hosting capacity, voltage stability, and peak-load mitigation, frequently relying on metaheuristic optimization to jointly site and size stations under electrical constraints [5]. While they improve technical feasibility, they can yield a poor user experience when travel behavior is not modeled in sufficient detail. The placement of charging stations is often motivated not only by charging demand and by economic advantages, but also by repercussions for the electrical network [14]. While a reasonable amount of studies have been conducted on the siting of charging stations, there are only few that consider working with large-scale eletric vehicle charging infrastructure [15].

2.3 Coupled and Bidirectional Approaches

A smaller but growing body of work couples transportation simulation with power-system analysis, for example through co-simulation frameworks that pass simulated charging loads

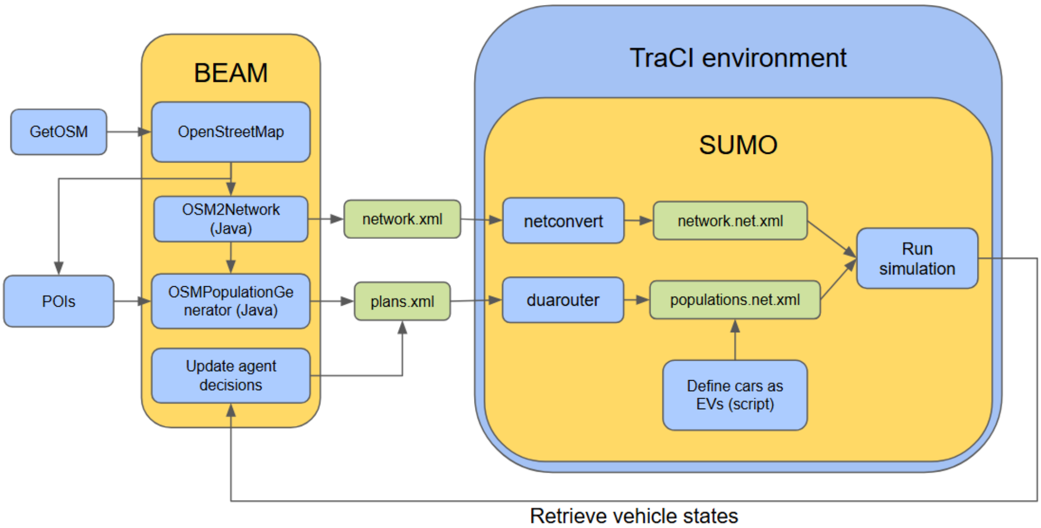

Figure 1: Data Pipeline under BEAM and MATSim.

to a distribution-grid model [10]. For bidirectional charging, planning models increasingly distinguish public and private chargers and account for uncertainty in demand and renewable generation [9]. Integrated and openly reproducible frameworks that combine microscopic mobility simulation with bidirectional charging at the level of individual public stations and private home wallboxes, however, remain limited. This paper addresses that gap on the mobility-and-demand side, while leaving a detailed electrical model of the distribution grid to future work.

3. Methodology

Initially, we chose MATSim for its high configurability and its ability to simulate large-scale traffic networks. To increase the realism of the simulation, we then incorporated BEAM, an extension of MATSim. The pipeline begins by fetching data from OpenStreetMap, which we utilized to generate the simulation network, including population, traffic nodes with links and dynamic agent decisions. After the workflow in MATSim and BEAM is completed, `.xml` files are generated for usage in SUMO and TraCI. With the *netconvert* module, the `network.xml` is converted to a SUMO-compatible `network.net.xml` and the population file is handled by the duarouter module. Once the population file was generated, a script injected EVs into it. As a last step, the simulation is run and the results are analyzed and fed back to update the agents iteratively (fig. 1). BEAM, which allows for a high grade of manipulation, is then used to generate files for traffic flow and charging behavior. However, its non-deterministic execution, high error rates, and extensive calibration requirements proved to be a barrier to productivity and reproducibility. We therefore switched completely to SUMO and TraCI, which provide a more stable and deterministic environment while still allowing detailed, dynamic modeling of traffic flow and charging behavior.

We use SUMO as the microscopic traffic simulator and TraCI for real-time control, combined with a grid-impact model and a statistical distribution of home-charging devices, to evaluate the spatial distribution of bidirectional charging stations. Two algorithms are implemented to test different siting strategies:

- **Saturate-and-Prune**: A naive iterative approach where candidate stations are placed on all viable road segments, ensuring maximum accessibility. After each trial execution, charging stations with less than a 5% utilization rate are removed, and the simulation is rerun until all remaining stations exceed this threshold. A 5% threshold was chosen to ensure that stations are neither underutilized nor overburdened, while also allowing for a sufficient number of stations to be placed. This approach is designed to maximize acessibility and coverage, but may result in a higher number of stations being placed than necessary, leading to potential underutilization of some stations.

- **Clustering**: A heuristic that prioritizes locations with high traffic flow, potential traffic congestion, and consideration for vehicles that enter a state of low State of Charge (SoC). Placement follows the traffic flow, aiming to maximize accessibility where vehicles are most likely to need charging while reducing detours.

3.1 System Overview

The fig. 2 illustrates the complete data pipeline, which begins in the same way as the BEAM pipeline, but diverges after the generation of the `.xml` files. The `chargingstation.xml` is utilized not only for the generation of the synthetic power grid, but also in assistance with the POI file to generate trip chains. To finalize the preprocessing, user-set parameters are applied to the mechanisms of the trip chain, private wallbox, and power grid generation. When preprocessing is complete, SUMO then uses the generated `trips.xml`, `wallboxes.xml` and `powergrid.pkl` files to simulate traffic flow and charging behavior. The simulation consists of three main components: TraCI, SUMO and PandaPower. The `powergrid.pkl` file is used with PandaPower for the grid-impact and the grid capacity determination, while the `wallboxes.xml` file is used to simulate the behavior of private wallboxes and the `trips.xml` file is utilized to run the individual agents with their trips. SUMO is the foundation that runs the simulation, while TraCI is used to retreive and manipulate values, routing of vehicles, detect charging stations and control wallbox charging. After a simulation run, the results are written to `.json` files and a `chargingsessions.csv` – which will be used to analyze the occured events, like low-SoC events, charging sessions, charging demand, traffic patterns and visualization of the simulation (fig. 3). If the custom web interface is accessible, the simulation will either be visualized directly (fig. 4) or the results can be accessed through the simulation results page.

3.2 Charging Station Placement Algorithm

3.3 Evaluation Metrics

4. Experiments

4.1 Experimental Setup

4.2 Datasets

4.3 Evaluation Metrics

4.4 Results

5. Discussion

5.1 Limitations

5.2 Interpretation of Results

5.3 Practical Example

6. Conclusion

This paper presented a simulation-driven method for distributing bidirectional charging stations under joint traffic and grid criteria. The evaluation demonstrates that coordinated planning can improve both charging accessibility and electrical reliability.

6.1 Future Work

Future work will include stochastic optimization under forecast uncertainty and integration of dynamic pricing signals. We also plan to validate the framework with larger real-world datasets and pilot deployments in mixed urban-suburban regions.

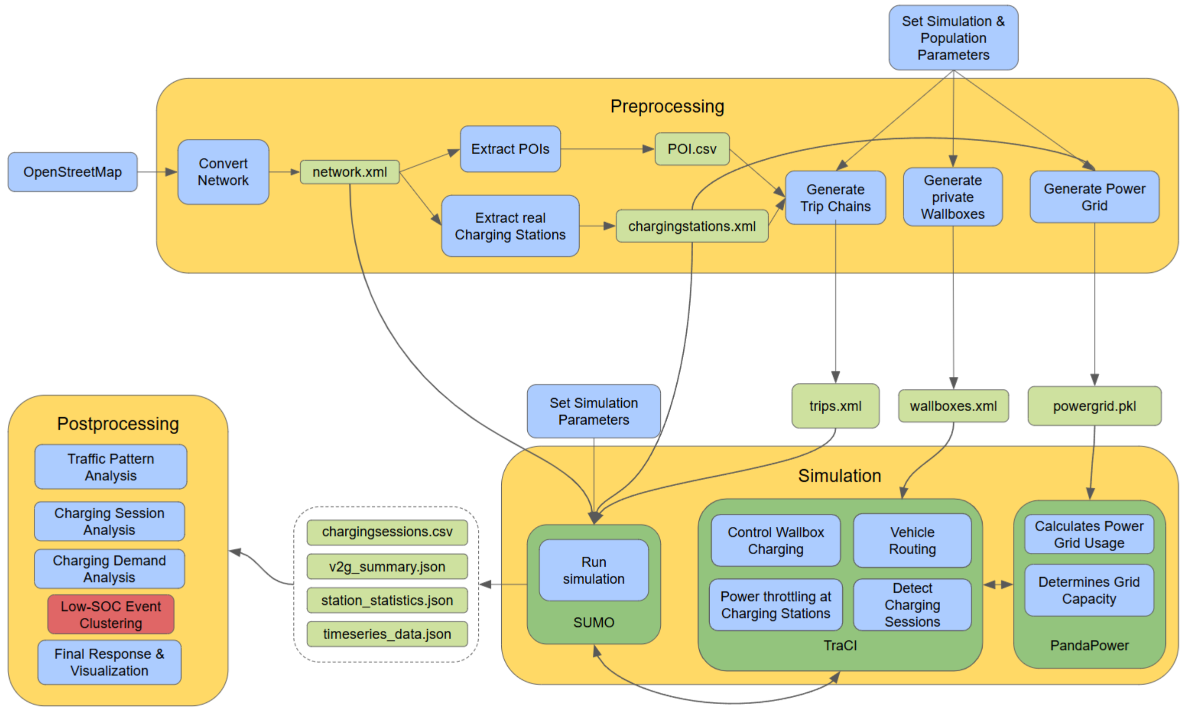

Figure 2: Complete Data Pipeline

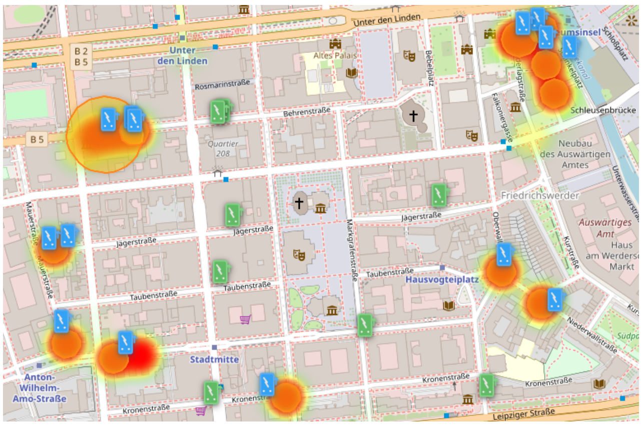

Figure 3: Resulting charging stations generated for the example run.

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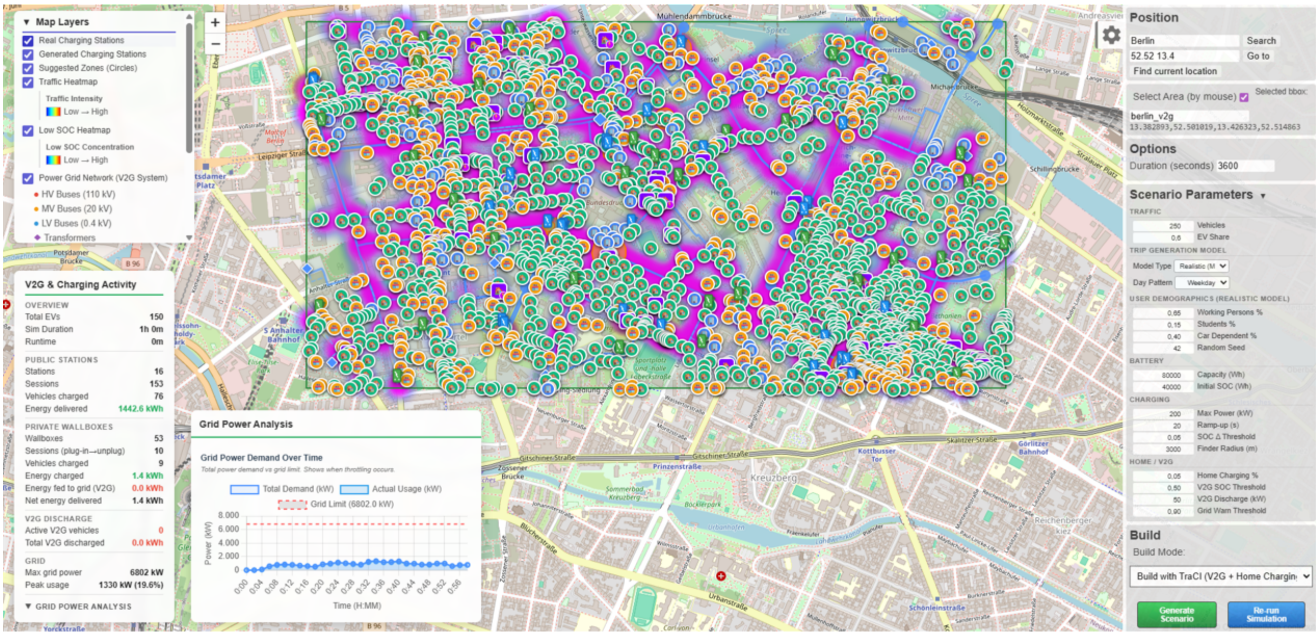

Figure 4: Example run of the workflow: the web interface with all data layers enabled.

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